

Array Decomposition

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TBD

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1 Introduction

2 Decomposition

2.1 Vectors

Suppose x is an element of n -dimensional Euclidean space, with the usual inner product and norm. We use a $\bullet$ replacing a subscript to indicate averaging over that subscript. We can decompose x into two orthogonal parts, with the first part a vector with all elements equal to the mean and the second part a vector with the deviations from the mean. Thus

$$x_i = x_\bullet + (x_i - x_\bullet). \quad (1)$$

Equation (1) can be written as

$$x = Px + (I - P)x, \quad (2)$$

where P projects on the subspace of all constant vectors, i.e. $P := n^{-1}ee'$ with e a vector with all elements equal to one. Now $P^2 = P$ and thus $P(I - P) = 0$, which shows that Px and $(I - P)x$ are orthogonal. Also, as Pythagoras used to say,

$$\|x\|^2 = \|Px\|^2 + \|I - Px\|^2. \quad (3)$$

The projection formulation suggests some obvious generalizations. We can replace the one-dimensional subspace of all constant vectors by an arbitrary subspace S , and we can replace the inner product $x'y$ by an arbitrary inner product $x' Ay$, with A positive semi-definite. If Y is an A -orthonormal basis for the subspace S , i.e. $Y'AY = I$, then $Px = YY'Ax$.

A further generalization uses multiple mutually orthogonal subspaces $S_1, \dots, S_m$, with corresponding projectors $P_1, \dots, P_m$. If $P_1 + \dots + P_m = I$ we can write

$$x = P_1x + P_2x + \dots P_mx, \quad (4)$$

and if $P_1 + \dots + P_m \preceq I$ in the Loewner sense then

$$x = P_1x + P_2x + \dots P_mx + (I - P_1 - \dots - P_m)x \quad (5)$$

Note that the “residual” $I - P_1 - \dots - P_m$ is orthogonal to all $P_1, \dots, P_m$.

2.2 Matrices

Suppose X is an $n \times m$ matrix, an element of the space $\mathbb{R}^{n \times m}$ with inner product $\text{tr } X'Y$ and squared norm $\|X\|^2 = \text{tr } X'X$. The decomposition we are interested in is

$$x_{ij} = x_{\bullet\bullet} + (x_{i\bullet} - x_{\bullet\bullet}) + (x_{\bullet j} - x_{\bullet\bullet}) + (x_{ij} - x_{i\bullet} - x_{\bullet j} + x_{\bullet\bullet}). \quad (6)$$

If we define the projectors

$$P_r = n^{-1}ee', \quad (7a)$$

$$Q_r = I - P_r, \quad (7b)$$

and

$$P_c = m^{-1}ee', \quad (7c)$$

$$Q_c = I - P_c. \quad (7d)$$

then equation (6) becomes

$$X = P_rXP_c + P_rXQ_c + Q_rXP_c + Q_rXQ_c. \quad (8)$$

Thus clearly shows four projectors, orthogonal to each other, adding up to the identity.

2.3 Arrays

$$\begin{aligned} x_{ijk} = & x_{\bullet\bullet\bullet} + (x_{i\bullet\bullet} - x_{\bullet\bullet\bullet}) + (x_{\bullet j\bullet} - x_{\bullet\bullet\bullet}) + (x_{\bullet\bullet k} - x_{\bullet\bullet\bullet}) + \\ & (x_{ij\bullet} - x_{i\bullet\bullet} - x_{\bullet j\bullet} + x_{\bullet\bullet\bullet}) + \\ & (x_{i\bullet k} - x_{i\bullet\bullet} - x_{\bullet\bullet k} + x_{\bullet\bullet\bullet}) + \\ & (x_{\bullet jk} - x_{\bullet j\bullet} - x_{\bullet\bullet k} + x_{\bullet\bullet\bullet}) + \\ & (x_{ijk} - x_{ij\bullet} - x_{i\bullet k} - x_{\bullet jk} + x_{i\bullet\bullet} + x_{\bullet j\bullet} + x_{\bullet\bullet k} - x_{\bullet\bullet\bullet}) \end{aligned} \quad (9)$$

3 Projectors

$$\{P_1(A)\}_{ijk} = x_{i\bullet\bullet}$$

$$\{Q_1(A)\}_{ijk} = \delta^{ijk} - x_{i\bullet\bullet}$$

$$\{P_{12}(A)\}_{ijk} = x_{ij\bullet}$$

4 Least Squares

5 Interactions

6 Log-linear Analysis

7 Orthogonal functions

Suppose M_n is the set of matrices X of order n with

- first column all elements equal to one, and
- $X'X = nI$.

Suppose the array A is $n \times m \times \ell$. Choose $X \in M_n$, $Y \in M_m$, and $Z \in M_\ell$. Define

$$\alpha_{ijk} = \sum_{i'=1}^n \sum_{j'=1}^m \sum_{k'=1}^l x_{i'j'k'} x_{i'i} y_{j'j} z_{k'k}$$